

Back to Square One

Foundations of AI

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How this works

- Nine questions. Some are multiple choice: put a hand up.
- The rest you **write down yourself**, on paper or on a laptop, and then we fill them in together on the screen.
- Everything on these slides was on a slide today.

A question the room gets wrong is worth five if it gets right, so guess out loud and be wrong cheerfully. That has been the whole day.

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Section 1

The loop

1. Write the algorithm

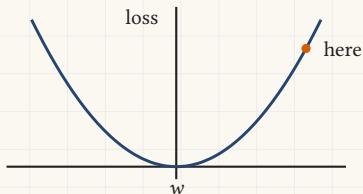
Block 1, one house. x is its size in square metres, y the price it sold for, and w the price per square metre we are looking for. The model is $y = w * x$.

```
def fit(x, y, w, alpha):  
    ----- # the verdict: y minus the estimate  
    ----- # while more than half a franc out  
    ----- # the update: move w the way the miss points  
    ----- # score the new guess  
    return w
```

Your turn. Write the four lines, then we fill them in together.

2. The tilt, on the side of the valley

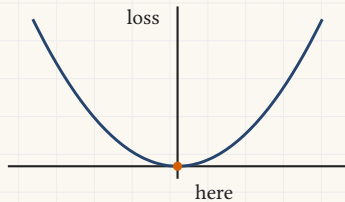
The curve is the loss. The dot is where the guess is now. What is the gradient there?



- A) about -12
- B) exactly 0
- C) about 0.1
- D) about $+12$

3. The tilt, at the bottom

Same curve. The dot has moved.



- A) about -12
- B) exactly 0
- C) about $+0.5$
- D) about $+12$

4. Which of these can draw any shape at all?

Given enough of them. More than one answer is right.

- A) Neural networks
- B) Straight lines
- C) Polynomials
- D) Loss functions

5. The network inside a language model

You type half a sentence and press enter. What does the network itself hand back?

- A) A word
- B) A matrix of words
- C) One probability for every word it knows
- D) The word with the highest score

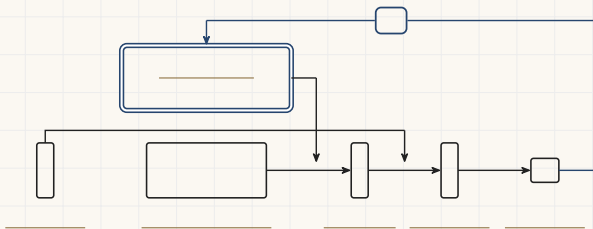
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Section 2

Name the boxes

6. The same five boxes: house prices

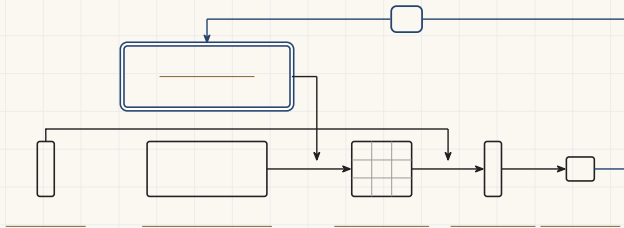
Every application today was this one picture. Here it is a network that predicts house prices.



Your turn. Name all six. Write them down, then call them out and I will fill them in.

7. The same five boxes: a language model

Nothing about the picture changes. Only what is in the boxes. One row is a piece of text with its next word cut off.



Your turn. The same six. One box is drawn wider than it was a moment ago. Ask yourself why.

8. That wider box, on its own

Two short texts, *the cat sat* and *the dog ran*. The five words along the top are everything this toy model knows.

<i>the</i>	<i>cat</i>	<i>sat</i>	<i>dog</i>	<i>ran</i>

after *the*, the next word was *cat*

after *the cat*, it was *sat*

after *the dog*, it was *ran*

Your turn. Three things. What is one **row**? What is one **column**? And if the model were perfect, what would be in every **cell**?

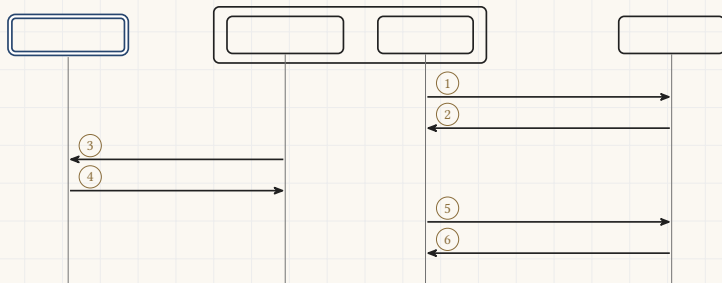
3

Section 3

The agent

9. What travels along each arrow?

The four actors of this afternoon, and the six messages between them, in order.



Your turn. One short line per arrow. The four boxes are worth naming too.

That is the day

- A guess, a measurement of how wrong it was, and a correction. Nine o'clock to five.
- The same five boxes behind a house price and behind a sentence.
- The only honest number comes from data the model never saw.
- A model writes text. A harness does things. Govern the harness.

Your project is on the course page: about an hour, in the browser, and it builds the house price network you met at question 6. Thank you for a Saturday.